

Exercise 1: Implementing the Singleton Pattern

Logger.java

```
class Logger
    private static Logger instance
    private Logger() {
        System.out.println("Logger Created");

    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;

    public void log(String msg) {
        System.out.println("LOG: " + msg);
    }
}
```

SingletonTest.java

```
public class SingletonTest
    public static void main(String[] args)
        Logger logger1 = Logger.getInstance();
        Logger logger2 = Logger.getInstance();
        logger1.log("Application Started");
        if (logger1 == logger2) {
            System.out.println("Only One Instance Exists");
        }
    }
}
```

Output

```
Logger Created
LOG: Application Started
Only One Instance Exists
```

Exercise 2: Implementing the Factory Method Pattern

Document.java

```
interface Document {  
    void open();  
}
```

WordDocument.java

```
class WordDocument implements Document {  
    public void open() {  
        System.out.println("Word Document Opened");  
    }  
}
```

PdfDocument.java

```
class PdfDocument implements Document {  
    public void open() {  
        System.out.println("PDF Document Opened");  
    }  
}
```

ExcelDocument.java

```
class ExcelDocument implements Document {  
    public void open() {  
        System.out.println("Excel Document Opened");  
    }  
}
```

DocumentFactory.java

```
abstract class DocumentFactory {  
    abstract Document createDocument();  
}
```

Factories

```
class WordFactory extends DocumentFactory {  
    Document createDocument() {  
        return new WordDocument();  
    }  
}
```

```
    }  
}  
class PdfFactory extends DocumentFactory {  
    Document createDocument() {  
        return new PdfDocument();  
    }  
}  
class ExcelFactory extends DocumentFactory {  
    Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

FactoryTest.java

```
public class FactoryTest {  
    public static void main(String[] args) {  
        DocumentFactory factory =  
            new PdfFactory()  
        Document doc =  
            factory.createDocument();  
        doc.open();  
    }  
}
```

Output

PDF Document Opened
